

RL Bootcamp 2026

Tutorial kick-off meeting

Initial ideas

- Start from existing code of previous tutorials
 - RL bootcamp 2024/2025
 - RL4AA 2024/2025
- We can use the following environments:
 - AWAKE beam steering task
 - Collision Avoidance in a flight simulator
 - Mujoco robotic tasks
 - RL QuantOpt

Fundamentals

Fundamentals

- Extensity.ai web examples
- tabular methods
- MC + TD(0) updates + formulae

- <https://rl-rover.extensity.ai/>
- <https://rl-grid.extensity.ai/>
- <https://rl-supply.extensity.ai/>
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Other idea for fundamentals

- Presentation of the maze RL lab from KTH (with agreement from the teachers since not publicly available)

Beginner

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- Colab notebooks introducing basic gym + SB3 RL training
- Swapping environments
- Swapping agents
- Basic transfer learning attempt
Train HealthyAnt - test on CrippledAnt

- https://colab.research.google.com/github/Stable-Baselines-Team/rl-colab-notebooks/blob/sb3/stable_baseline_s_getting_started.ipynb
- https://colab.research.google.com/github/Stable-Baselines-Team/rl-colab-notebooks/blob/sb3/monitor_training.ipynb
- Use the Colab introduction that I am writing, to be added to the fork in the GitHub repository :
https://colab.research.google.com/drive/19d4VPG7VluxwdXjLv_hcNwl6Gct6NrZ6?usp=sharing

Intermediate

→ Solve the CrippledBot (Co) problem

→ Train on Atank, Test on Htank

→ Increase injury levels

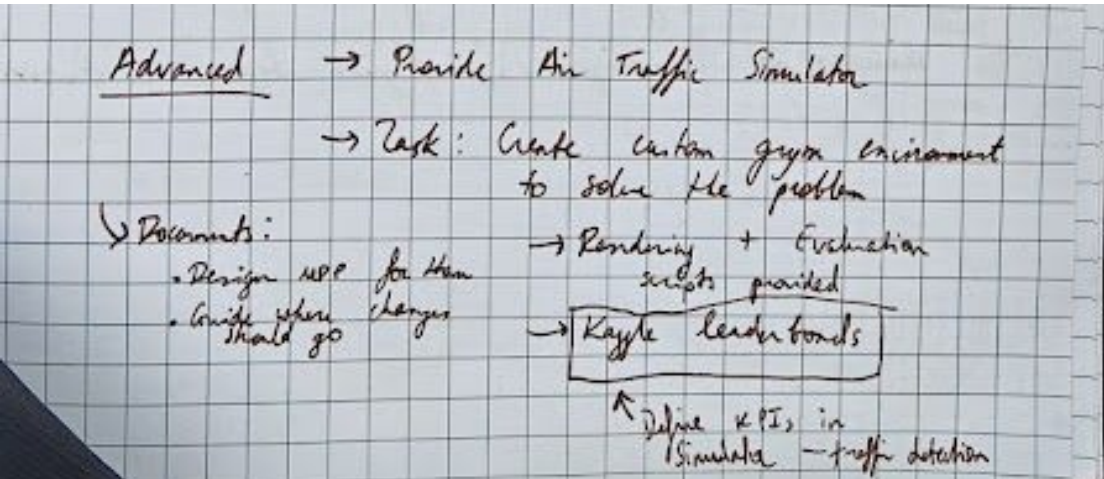
→ Use SB3 + Hydra

Intermediate

- We can start from an RL module taught in UoM (CCE5502)
 - <https://github.com/leander-grech/CCE5502-RL-Assignment-2025.git>

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Advanced



- <https://indico.kit.edu/event/3746/sessions/3777/>
- <https://github.com/RL4AA/rl4aa24-tutorial>
- https://github.com/MathPhysSim/Tools_for_controls
- Beam example and MDP comparison for quantum computing and example of the beam (rl-quant-opt)

Advanced

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